



SPI FLASHING TOOL

User's Guide

Table of Contents

Table of Contents.....	2
List of Figures.....	2
Version table.....	4
1. Overview	5
2. System Requirements	6
3. Flashing Tool on Linux.....	6
4. Flashing tool User Interface Overview.....	6
5. Using the GUI Tool to Flash	9
6. Using the Command line	23
7. Basic UART Test.....	28
Appendix	29

List of Figures

Figure 1: SPI Flashing Tool architecture	5
Figure 2: Flashing tool.....	5
Figure 3 : Flashing tool options.....	6
Figure 4 : GUI outlook	9
Figure 5 Selecting the board Type	9
Figure 6 Example .flash.cfg.xml.....	10
Figure 7: Browsing to select the files	10
Figure 8: Selecting the files to be transmitted.....	11
Figure 9: Selected file's path	11
Figure 10: SPI NOR flash address in hexadecimal	11
Figure 11: End address till which the file will be flashed	12
Figure 12: Flash address of 2 nd file overlaps with 1 st file	12
Figure 13: Error asking to mention address for 2 nd file.....	13
Figure 14: Correct way of mentioning addresses	13
Figure 15: Selecting SPI speed	13
Figure 16: Selecting SPI mode.....	14
Figure 17: Selecting COM port.....	14
Figure 18: Checking for baud rate switching	15
Figure 19: Baud rate switched to 1000000.....	16
Figure 20: Preparing flash for selected files.....	16
Figure 21: Flashing the file	17
Figure 22: Flashing Completed successfully.....	18

Figure 23:Checking for baud rate switching for chip erase	19
Figure 24: Performing chip erase.....	19
Figure 25: Chip erase complete	20
Figure 26: Reconnect button	20
Figure 27: Quitting the tool.....	21
Figure 28: Trying to close the tool during operation	22
Figure 29: flash_programming_cmdline.py usage.....	25
Figure 30: Checking baud rate switching	25
Figure 31: Prepare flash	26
Figure 32: Ongoing flash	26
Figure 33: Flashing completed	27
Figure 34: Asking user if he wants to continue with chip erase	27
Figure 35: Performing chip erase after “y” option.....	27
Figure 36:Chip erase done	27

Version table

Release	Date	Author	Comment
0.9	28/01/2022	Andhe Anil Kumar	• User guide for flash issi
0.91	10/02/2022	Andhe Anil Kumar	• Updated board name changes • Internal code change to check flash is in XIP mode. If it is in XIP mode, exit out of XIP mode and jump to usual flashing procedure.
1.0	22/02/2022	Andhe Anil Kumar	• Checksum added
1.1	09/10/2023	Ganesh Kumar R	• Support for RTSS Release v0.96 • Only Linux Support with Executable Binary.
1.2	07/07/2025	Ganesh Kumar R	• Eagle Devkit support

1. Overview

SPI Flashing Tool is a Host PC application developed for Linux (Ubuntu 20.04).

This utility will be used to transfer FW images from a Host PC to the Target System using UART & program the FW images to the ISSI Flash.

This document demonstrates how to use the SPI flashing tool in order to transfer the FW images to the flash.

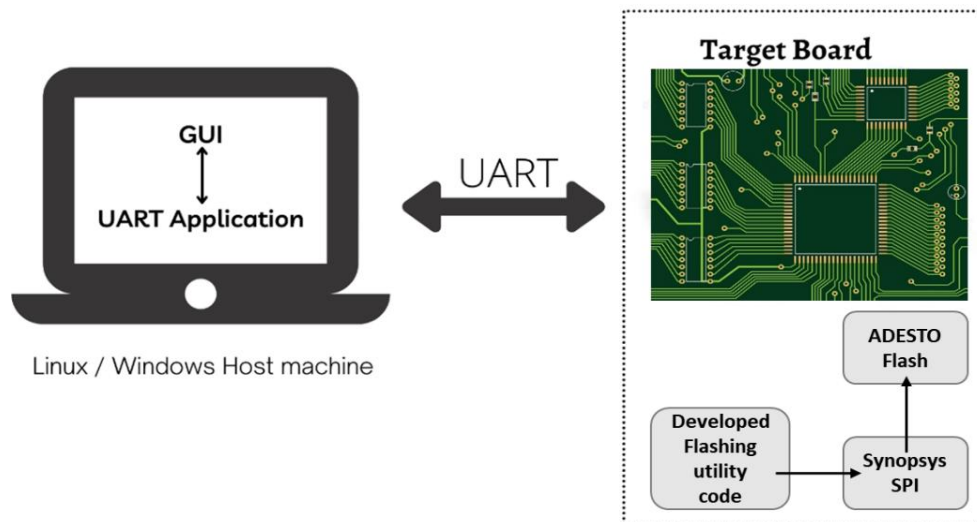


Figure 1: SPI Flashing Tool architecture

The flashing tool window is shown in Figure – 2.

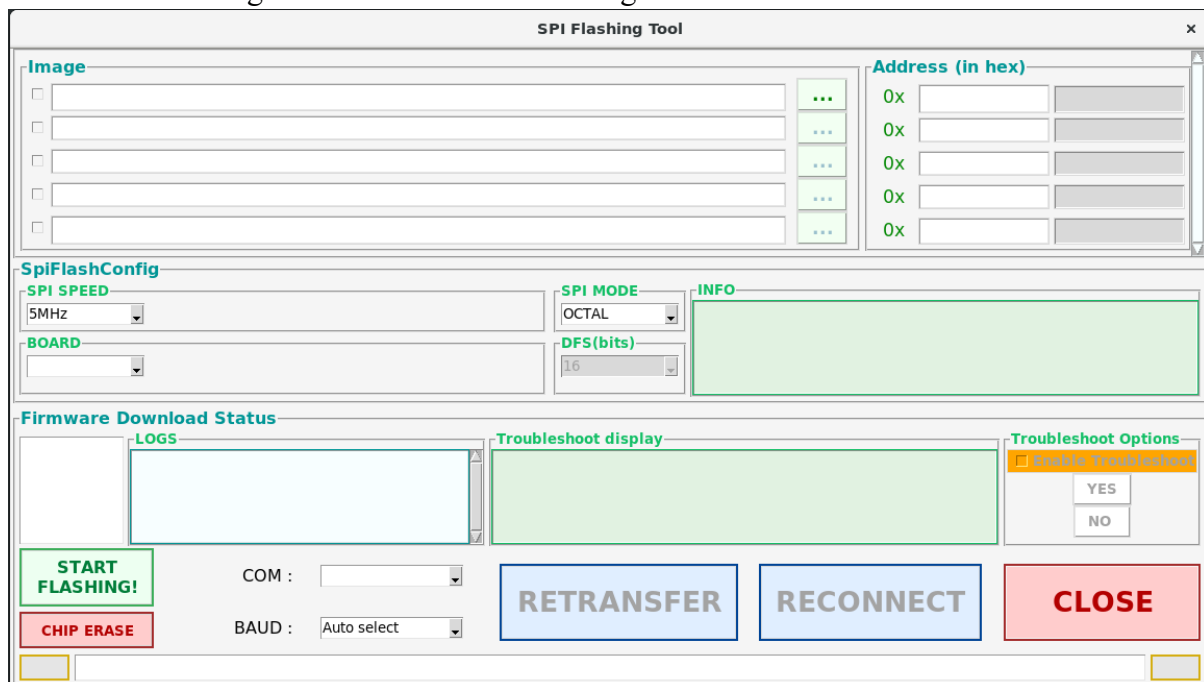


Figure 2: Flashing tool

2. System Requirements

- Target Board (Dev Kit) connected to UART 2.

3. Flashing Tool on Linux

This tool is a Linux Executable binary, downloads all the listed files and keep in a folder.

- FlashTool (Executable)
- Support Files
 - .flash.cfg.xml
 - alif_logo.png

4. Flashing tool User Interface Overview

The below figure-3 shows the options available in the flashing tool which will be discussed next.

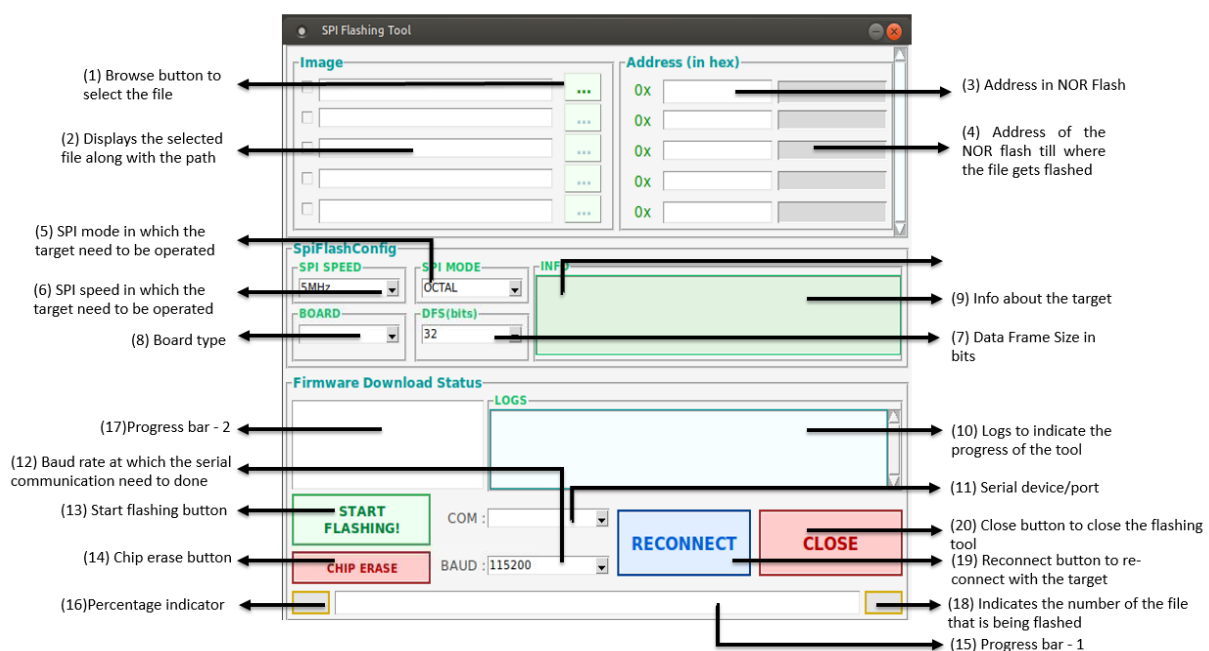


Figure 3 : Flashing tool options

1. File Browse Button:

Browse button helps to select the files which you want to transmit. Click on this button and select the file you wish to transfer.

2. Selected file:

The selected file path will appear in this box.

3. Address in SPI NOR flash:

The address at which the corresponding file need to be flashed is mentioned here in this box.

- Since the dfs is fixed as 32, then the given address should be divisible by 4

4. Address range:

This will show the end address in flash memory at which last byte will be written.

5. SPI Mode:

User can configure the target board SPI mode here

- OCTAL

6. SPI speed:

The clock frequency at which the target needs to be operated is selected from the drop-down box.

7. Data frame size in bits:

The data frame size format needs to be selected. The available options are –

- **32:** Data will be written in multiple of words. If the file size is not aligned with 4 bytes boundary. The last chunk of data will be padded with 0 to make it 4 byte aligned. For ex: if file size is 255, then one byte will be padded to make it 4 byte aligned.

8. Board type:

Select the board from the drop-down menu where carrier board and dev board options are available. Flashing tool will read information for selected board type and size from the “.flash.cfg.xml” file.

9. Info box:

Any information about the target such as chip id & run time updates will display here.

10. Logs box

Logs will display here.

11. Serial device/port

The port to which the target is connected need to be selected from the list of available ports. Serial communication will get established with the target via this port.

12. Baud rate:

Select the baud rate from drop down menu where 'Auto select' and '115200' are the two available options. If 'Auto select' is selected, the tool will try to switch to higher baud rate upto 1000000. The initial communication takes place at default 115200. If the higher baud rate is available the tool will negotiate with the target to use the higher baud rate. If the negotiation fails the tool will use the default baud rate, otherwise uses the negotiated baud rate.

13. Start flashing button:

Once everything is set, clicking on this button will start transferring the files to the target.

14. Chip erase button

This option let the user to erase the whole flash

15. Progress bar-1

The progress bar will give a visual illustration of the process completion time of the transmission of each file.

16. Percentage indicator

Indicates the percentage of file transferred.

17. Progress bar-2

This is similar to the progress bar-1, displays the completion time and percentage of the transmission of all the files that need to be transferred to the target.

18. File indicator

This bar shows the total files need to be transferred and the number of the current file being transferred.

19. Reconnect button

Once a flash is done, if the user wish to use the tool again, clicking on this button will reset the GUI. It closes the opened serial port and clears the selected options. Now, the user can use the tool to flash again.

20. Close button

Clicking on this button will close the opened serial port and closes the tool as well.

5. Using the GUI Tool to Flash

Run the provided flashing tool application by the following command.

For Linux: `$sudo ./flashtool`

When the above command is executed, a new window pops up on the screen.

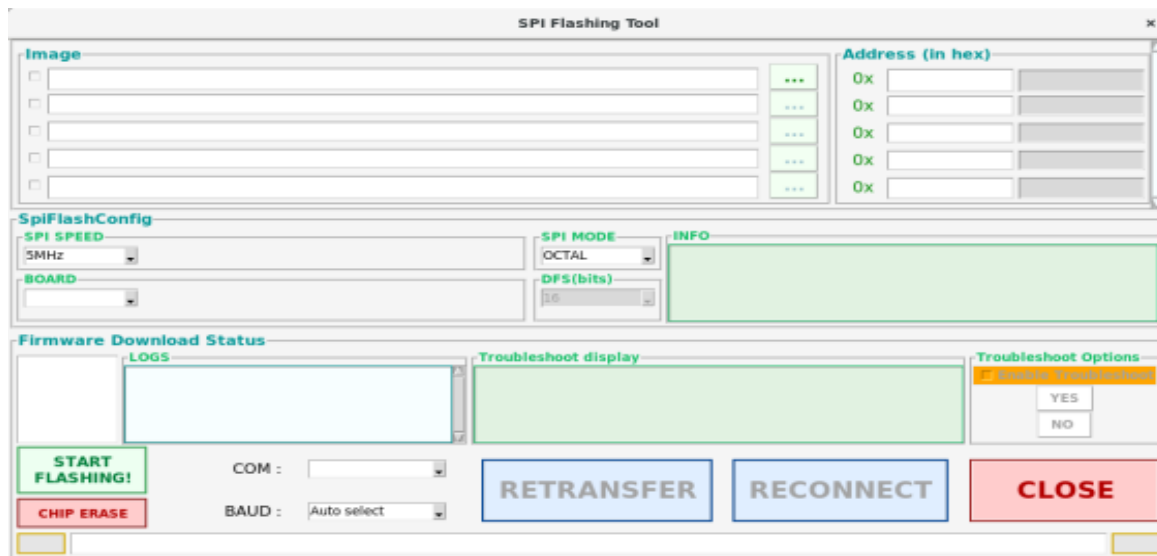


Figure 4 : GUI outlook

NOTE: Throughout this document, Linux screenshots are used.

5.1. Select the board type

From the SPI flash config section, under BOARD, select the board type from the available list. This list is fetched from the “.flash.xfg.xml” file.

Selecting the board helps in transferring the files without exceeding the corresponding flash size. User will get notified if the size of the selected file/files exceeds the size of the flash.

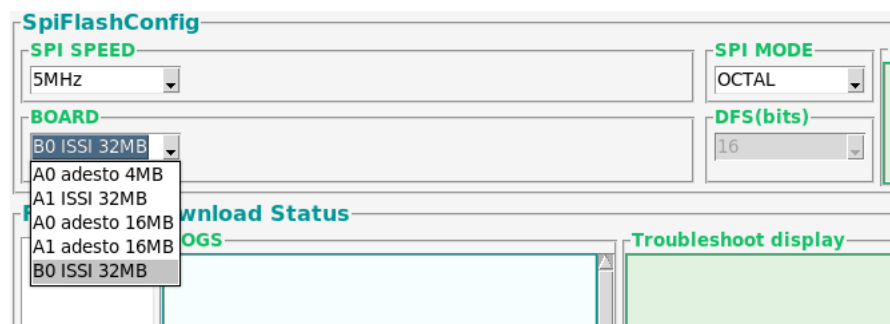


Figure 5 Selecting the board Type

An example of configuration file “.flash.cfg.xml” is shown in Figure-6.

User must use the board and flash details as per the below xml.

- User chosen board name.
- Flash name must be “issi” or “adesto” as per the flash present on the board. No other flash name is recognized.
- Start address of SPI memory map.
- The size must be mentioned in bytes as shown in Figure-6.

Note: User must strictly follow the example xml and should not add extra spaces before and after the board and flash name, start address and size.

```
<!-- .flash.cfg.xml -->

<head>

    <TARGET>
        <board>A0 adesto 16MB</board>
        <flash>adesto</flash>
        <start_address>0xD0000000</start_address>
        <size>16*1024*1024</size>
    </TARGET>

    <TARGET>
        <board>A1 adesto 16MB</board>
        <flash>adesto</flash>
        <start_address>0xD8000000</start_address>
        <size>16*1024*1024</size>
    </TARGET>

    <TARGET>
        <board>B0 ISSI 32MB</board>
        <flash>issi</flash>
        <start_address>0xC0000000</start_address>
        <size>32*1024*1024</size>
    </TARGET>

</head>
```

Figure 6 Example .flash.cfg.xml

5.2. Select the files

- Select the files that need to be transmitted to the target by clicking on the browse button (highlighted in the Figure-7).

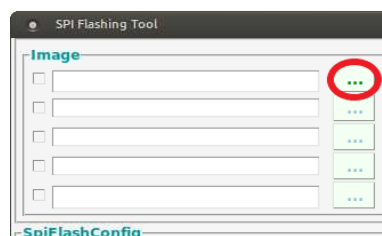


Figure 7: Browsing to select the files

- Browse window opens, user can now select the file. Only one file to be selected. To demonstrate, bash file is selected as shown in Figure-8. Once a file is selected, the selected file along with the path will get displayed in the tool as shown in the Figure-9.

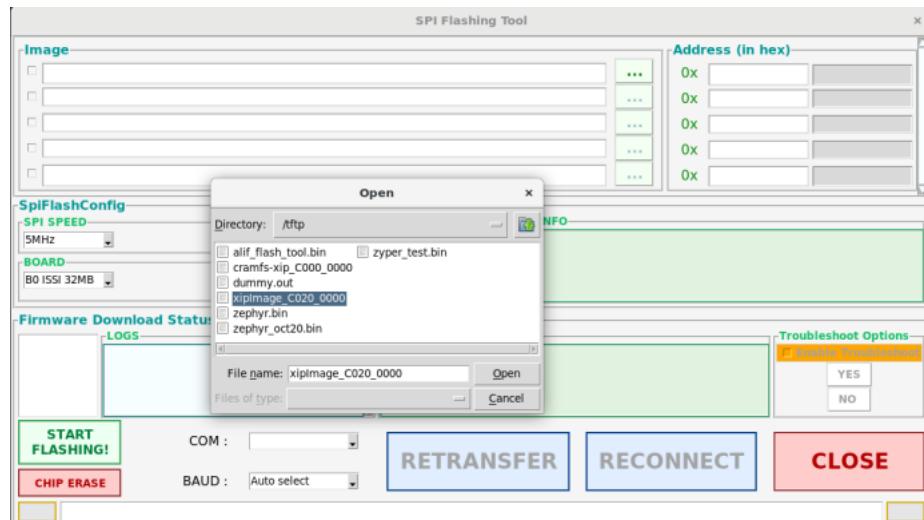


Figure 8: Selecting the files to be transmitted



Figure 9: Selected file's path

NOTE:

All the files which need to be flash must be selected together before start flashing. Writing file one after one will corrupt the previous written file, if file-1 end address and file-2 start address lies in the same $4KB \times n$ block (n is block number).

If file-1 end address is in $4KB \times n$ block and the file-2 is $4KB \times n+1$ block, then both files can be written successfully.

5.3. Mention the address

- After selecting the file, mention the SPI NOR flash address in hexadecimal in the address section at which the file has to be flashed. As shown in the Figure-10, the selected file (bash) will be flashed at XIP address $0xC0200000$.

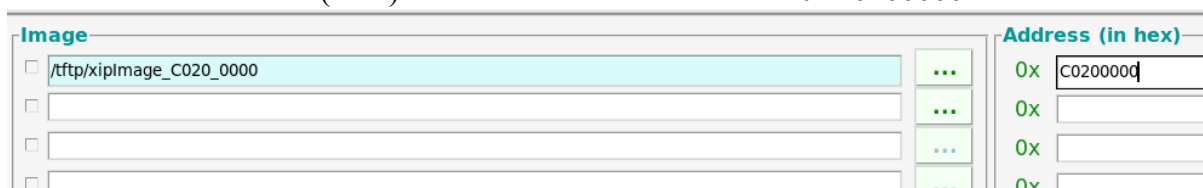


Figure 10: SPI NOR flash address in hexadecimal

- After mentioning the address, hit Enter (Return) key in the keyboard. Based on the size of the selected file, the end address till which the selected file will be flashed is calculated and gets displayed in the address section when the user hits Enter key, once address is mentioned.

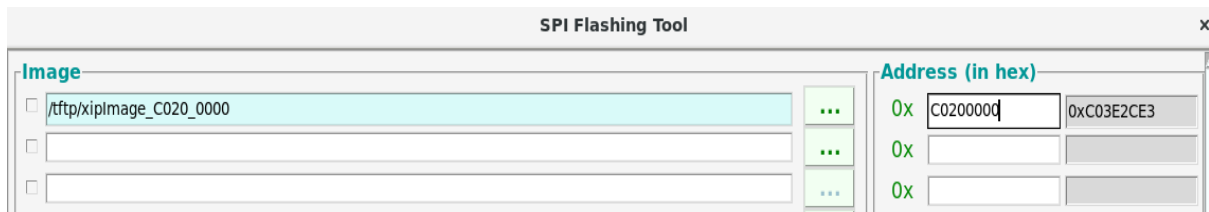


Figure 11: End address till which the file will be flashed

5.4. Flash multiple files

Click on the next browse button, select the file, and mention the address for the corresponding file. Follow the section 5.2.

NOTE:

1. The SPI NOR flash address for the next file should not overlap the previous file. If the user mistakenly specifies the address that overlaps the previous file, flashing tool gives a warning saying that file overlaps the previous file as shown in Figure-12.

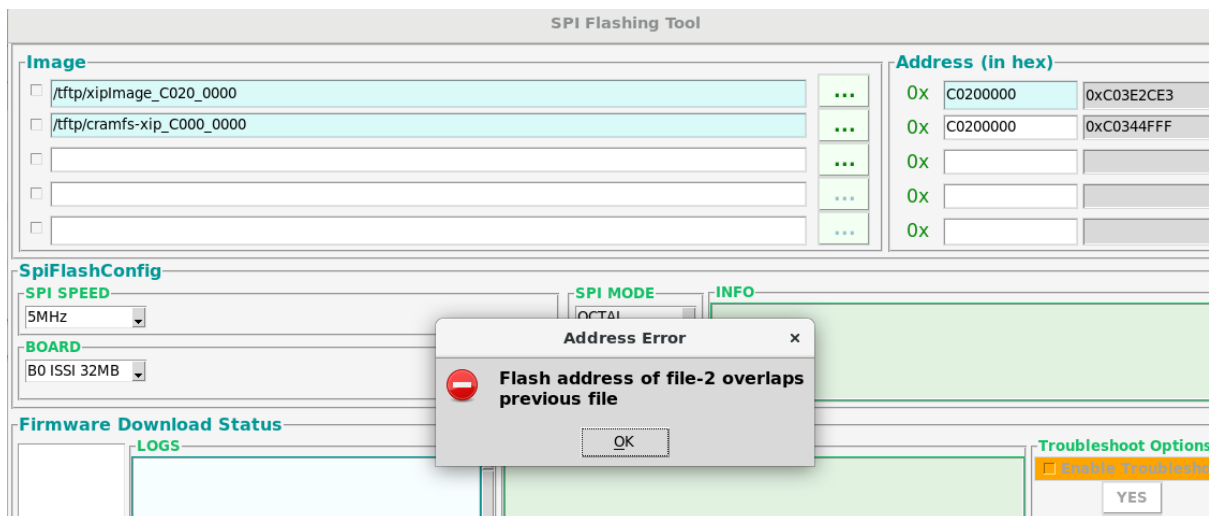


Figure 12: Flash address of 2nd file overlaps with 1st file

2. If the user tries to select next file by clicking on the browse button without mentioning the address for the previous file, the tool notifies to mention the address for previous file and then proceed. User will not be able to select next file unless the address for the previous file is mentioned.

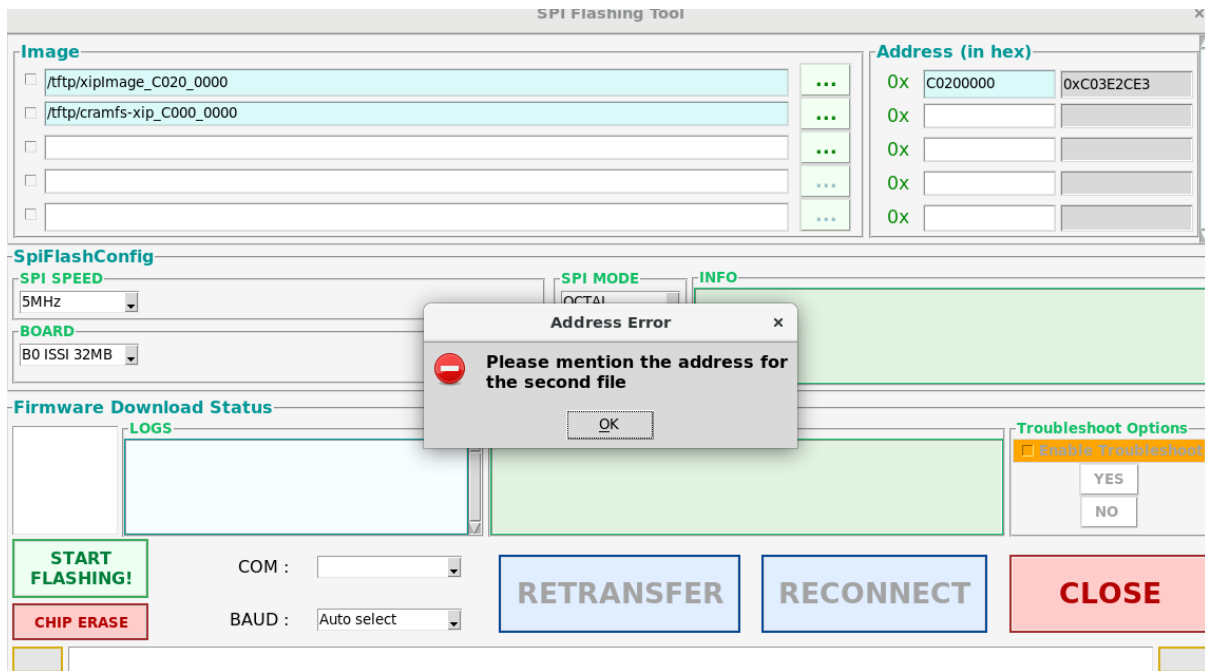


Figure 13: Error asking to mention address for 2nd file

If the address is mentioned correctly the tool will appear as shown in the Figure-14.

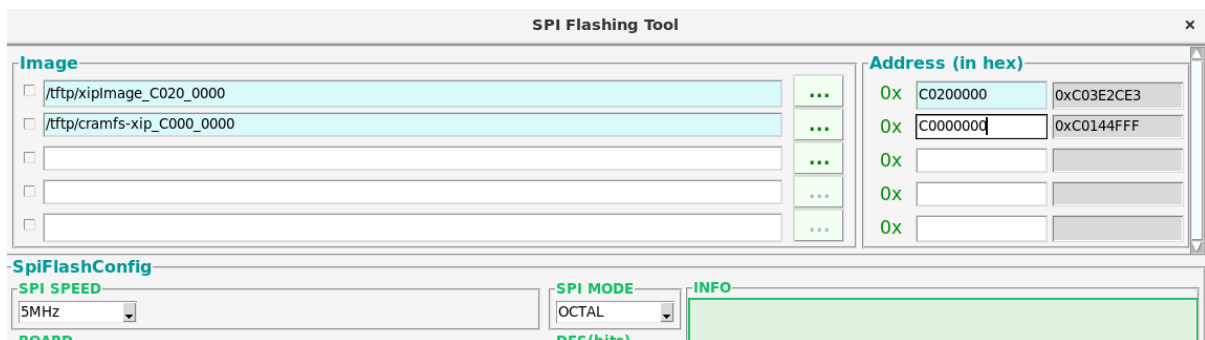


Figure 14: Correct way of mentioning addresses

5.5. Select SPI speed

Select the SPI speed in SPI flash config section.

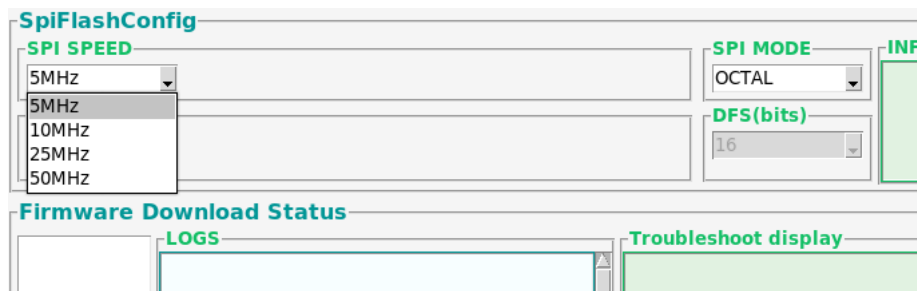


Figure 15: Selecting SPI speed

5.6. Select SPI mode

Select the SPI mode in SPI flash config section.

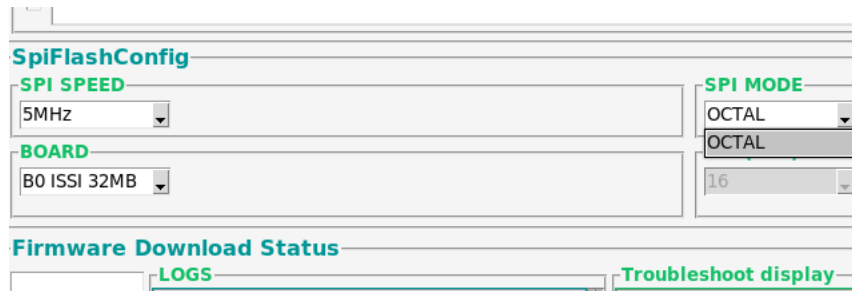


Figure 16: Selecting SPI mode

5.7. Select COM port

Select the COM port. Transfer of files will not be done if the port is not selected. Flashing tool will notify the user to select the port in case not selected.

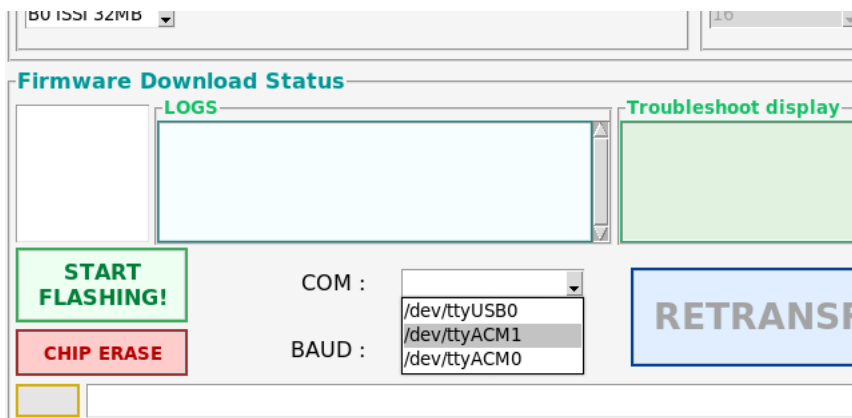


Figure 17: Selecting COM port

5.8. Start writing the Flash:

If everything is set, clicking on START FLASHING! Button, will start transferring files.

Initially, tool will check for the availability of high baud rate than the default baud rate. The serial port will be opened with default baud rate and start communicating with the target. If higher baud rate is available, the same is shared with the target. The tool will now switch to new baud rate and checks whether the new baud rate is supported by both target and host by communicating to each other using new baud rate. On successful communication with new baud rate, now the baud rate for the rest of the process will be set to the new baud rate, default baud rate is used otherwise.

Once the connection is established, the SPI flash config parameters such as board type, SPI speed and mode, data frame size will be shared with the target. Then the details of the files such as size, address will be shared with the target and waits for the acknowledgment from the target. During the wait time, the host will display preparing flash till target acknowledgement is received as shown in the Figure 21.

Once the acknowledgement from the target is received, transfer of files will begin and all the selected files will be transferred one after the other at the specified addresses. Every communication with the target will be displayed in the Logs box in the flashing tool and also recorded in the Logfile.txt file.

- Once the start flashing is pressed the tool will check for higher baud rate as shown in Figure-18.

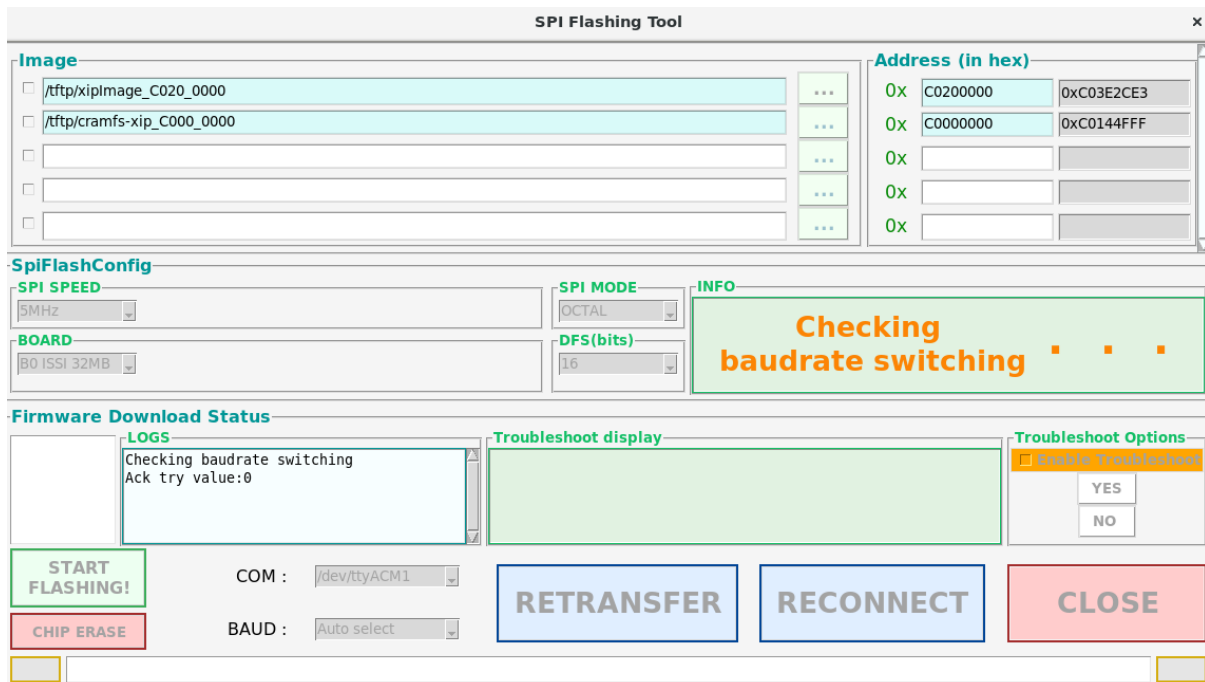


Figure 18: Checking for baud rate switching

- If higher baud rate is available and supported by both host and target, the baud rate is set to the higher value and the new baud rate is used for the rest of the process.

Note: For Eagle A0 Devkit please select baud rate to 115200.

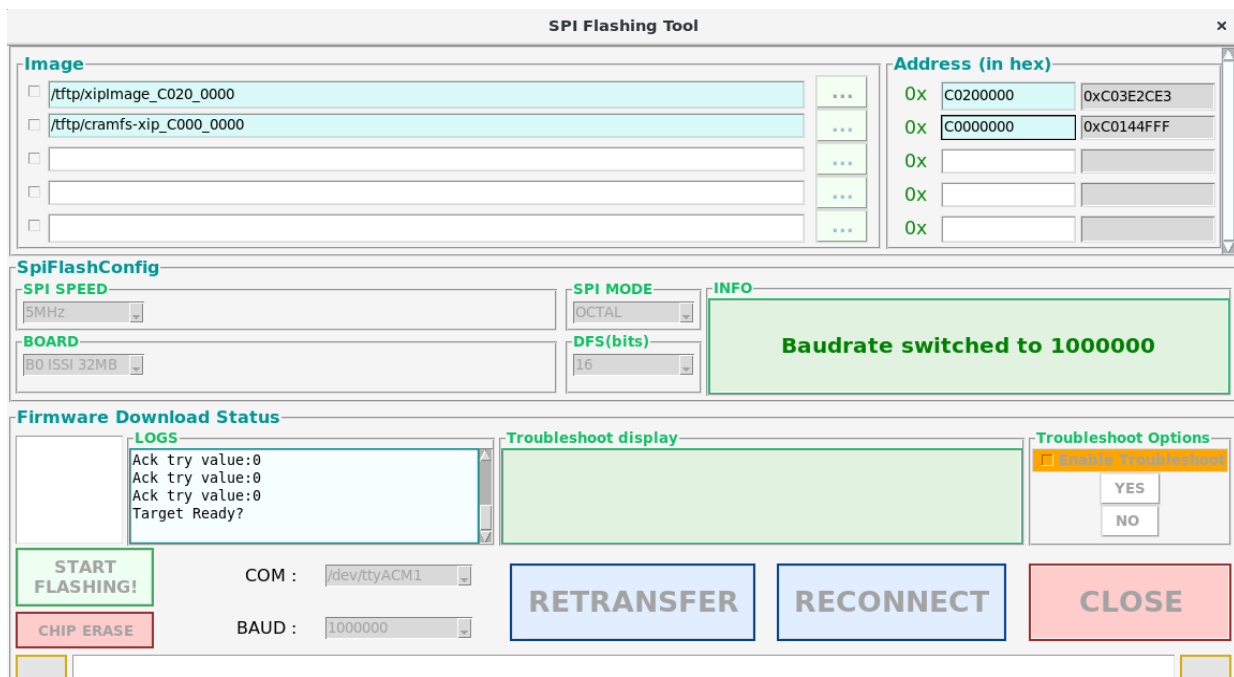


Figure 19: Baud rate switched to 1000000

- If there is no availability of high baud rate, default baud rate is used.

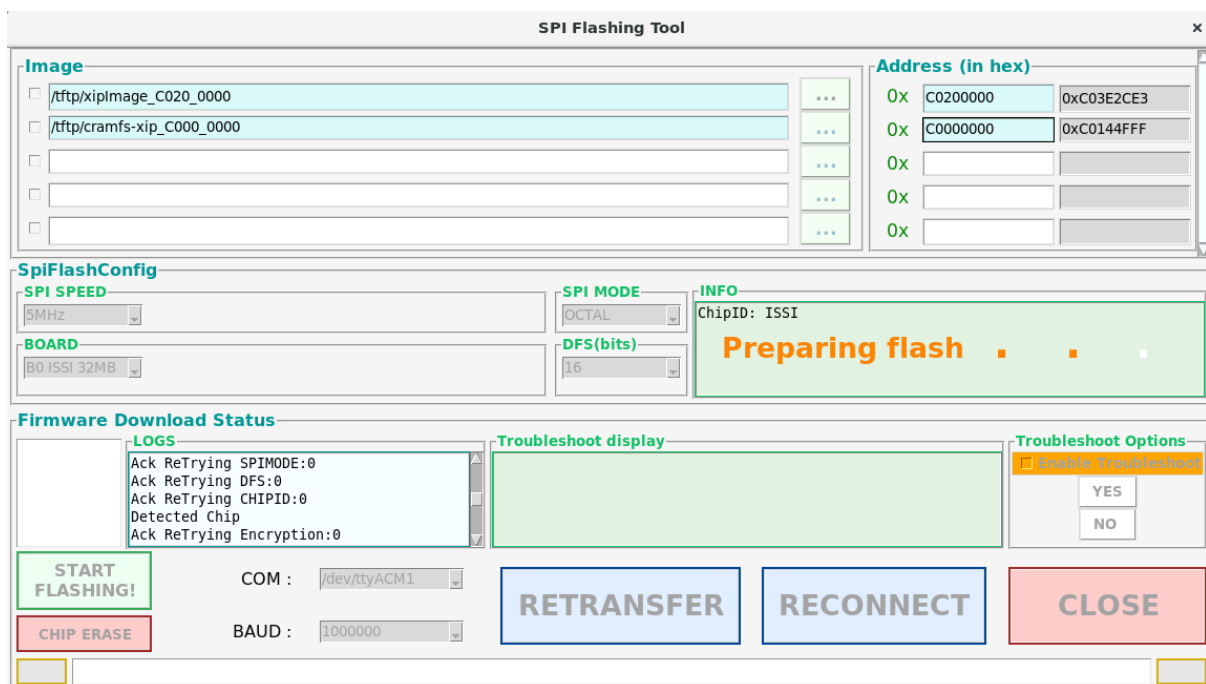


Figure 20: Preparing flash for selected files

- Program on Target Board will receive the SPI configuration parameters + FW image from Host Flashing Utility over UART interface. This program shall receive the FW image in blocks over UART & write these blocks to the Flash. The transfer completion time of a file can be seen in the progress bar-1 and the completion time to transfer all the selected files can be seen in the progress bar-2.

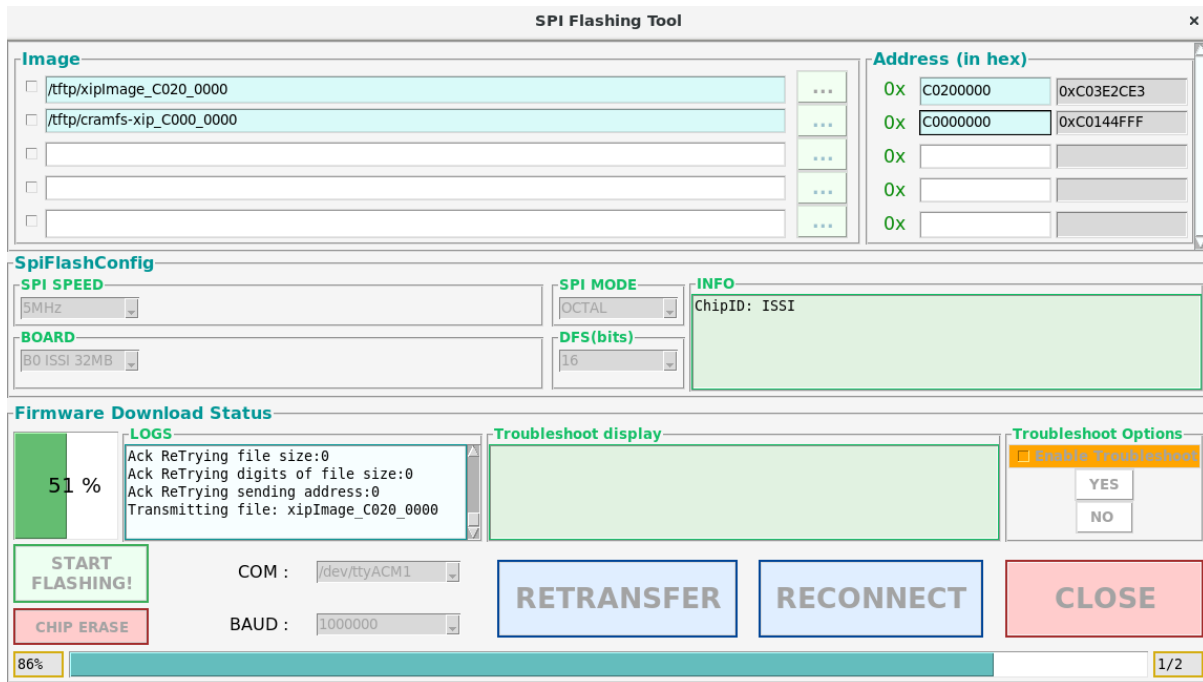


Figure 21: Flashing the file

After all the selected files are transferred, 'Finish' label will show up in the progress bar-2 and all the options are disabled except RECONNECT and CLOSE buttons as shown in Figure-23.

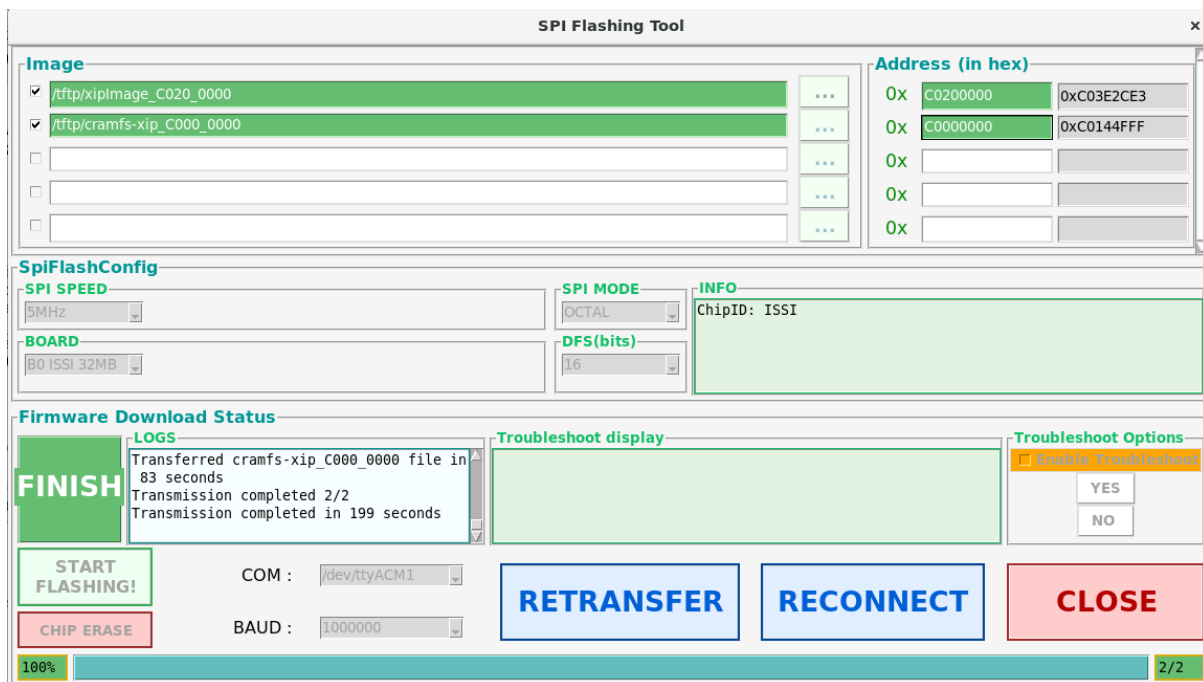


Figure 22: Flashing Completed successfully

5.9. Performing chip erase

Once the CHIP ERASE button is clicked, the tool will send chip erase message to the target. Then target will erase the whole flash and send the acknowledgement to the host. During the chip erase, the tool will display 'Performing chip erase' label as shown in figure-25. Once the acknowledgement for chip erase completion is received,

the tool will display ‘Chip erase done’. After the completion of chip erase all the options are disabled except RECONNECT and CLOSE buttons as shown in Figure-25.

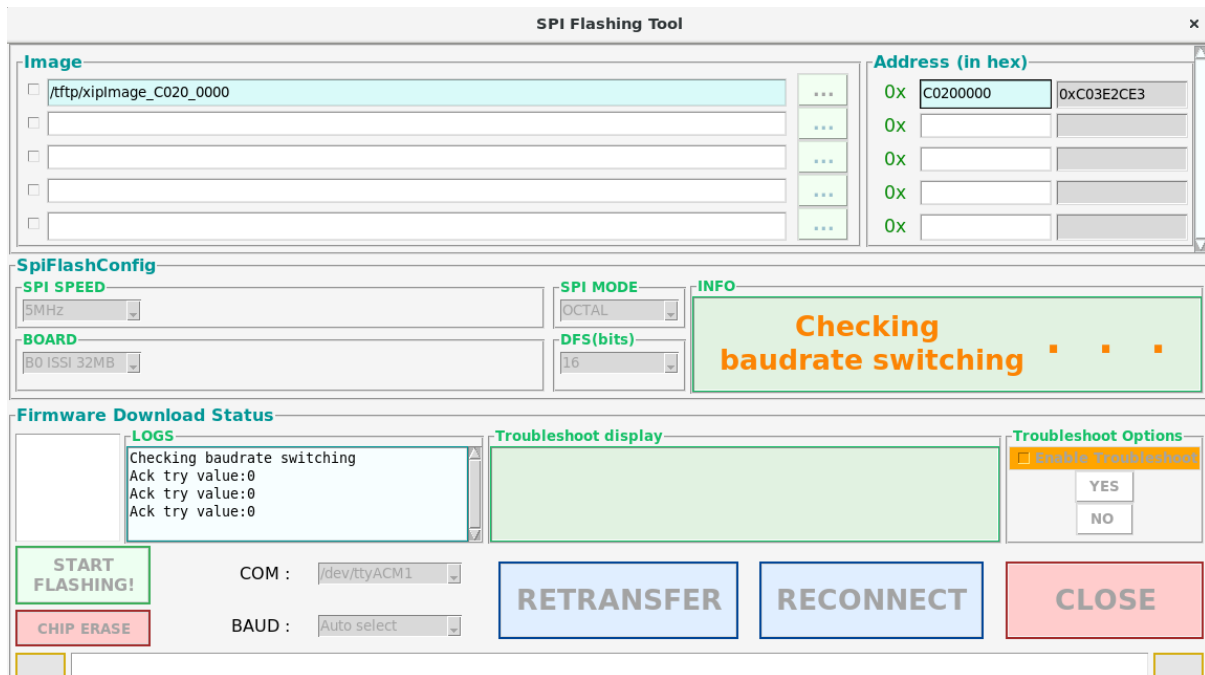


Figure 23: Checking for baud rate switching for chip erase

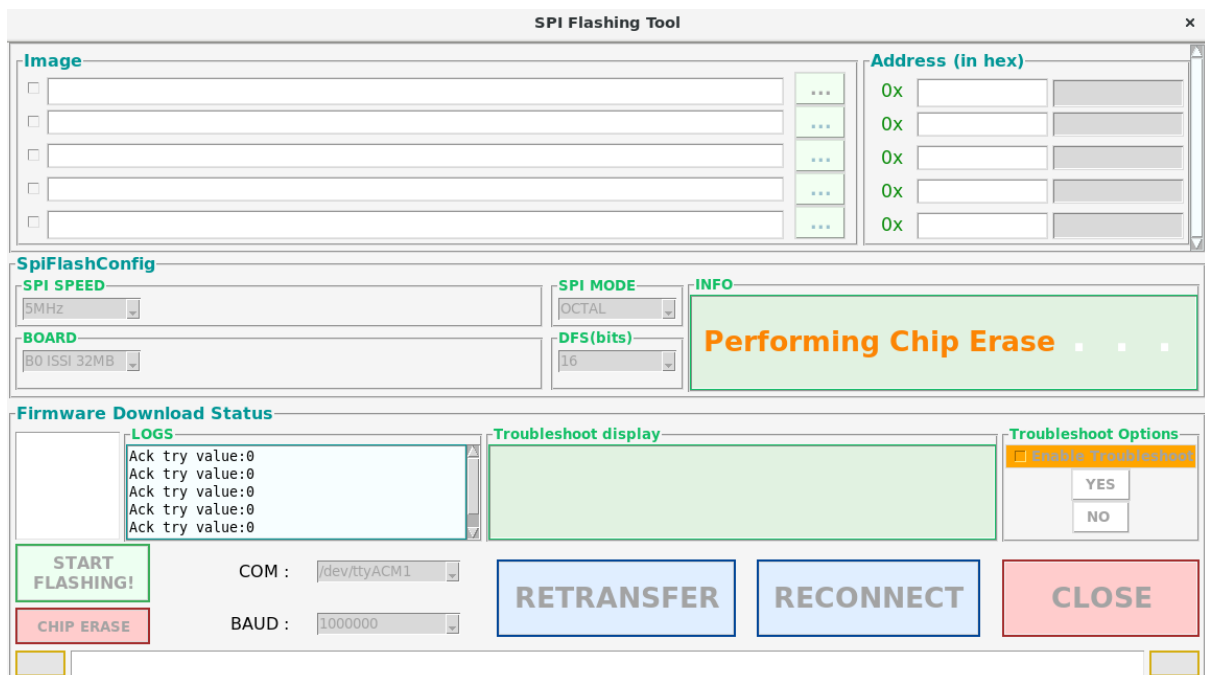


Figure 24: Performing chip erase

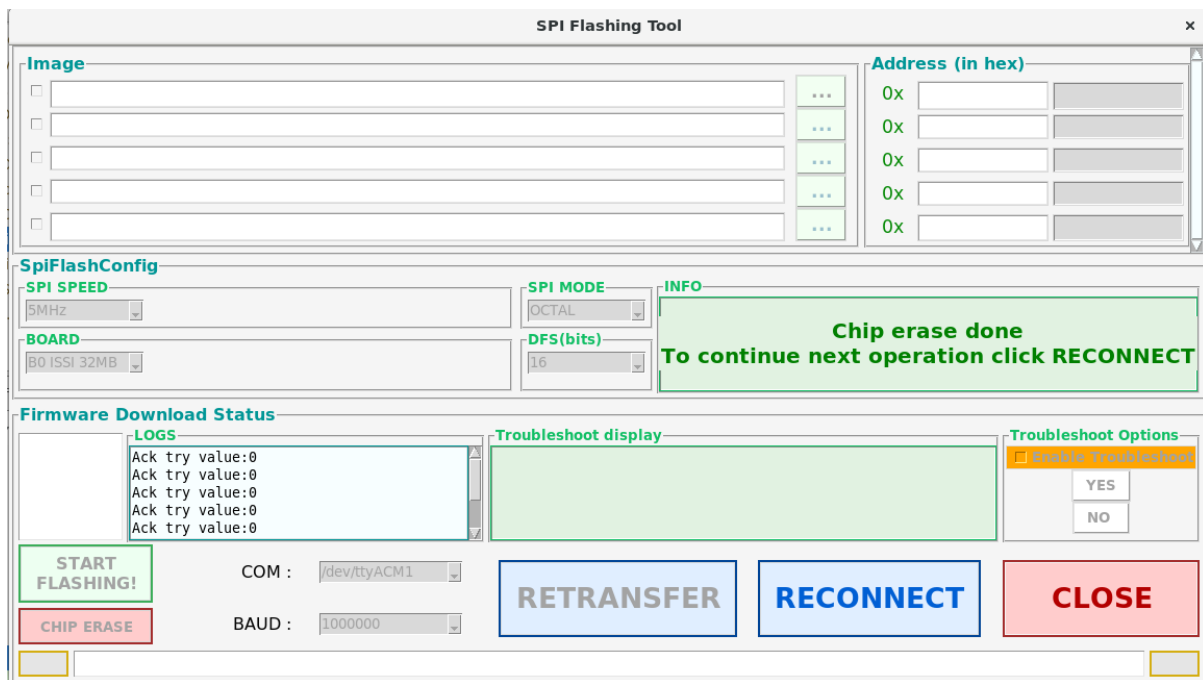


Figure 25: Chip erase complete

5.10. RECONNECT Feature

Once the “START FLASHING!” or “CHIP ERASE” operation completed, if the user wishes to use the flashing tool again, clicking on reconnect button will reset the tool and the user can select the files, SPI flash config parameters and start flashing again.

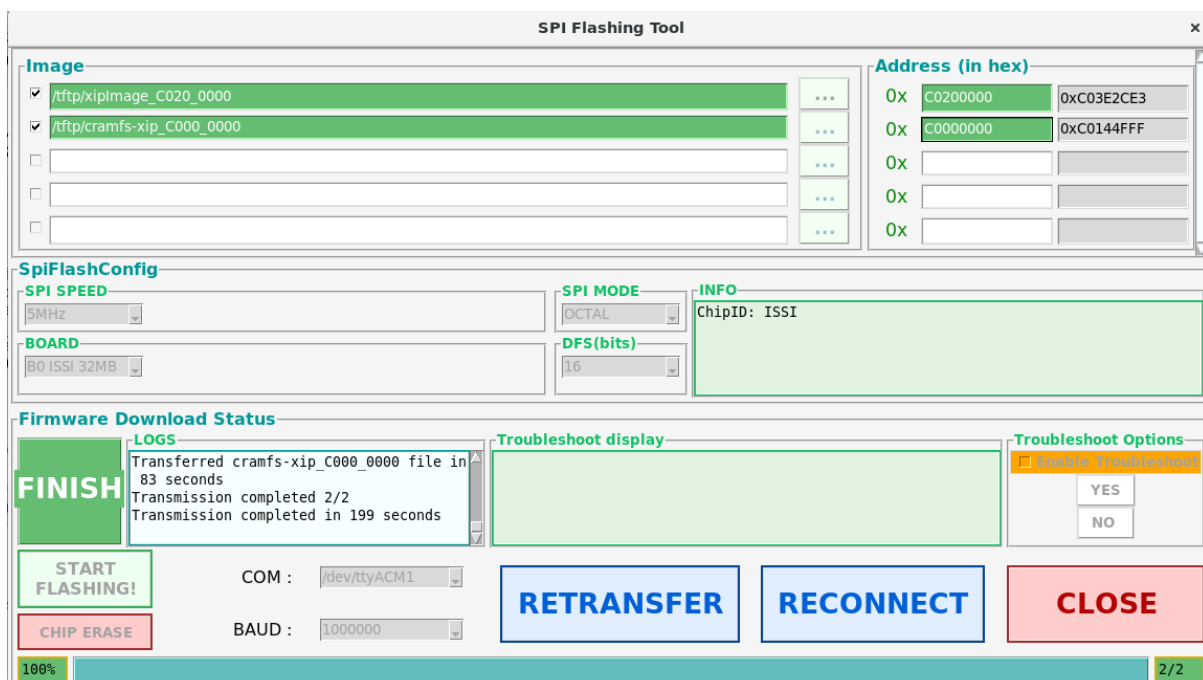


Figure 26: Reconnect button

NOTE: If reconnect feature is used, checking for higher baud rate will not happen again since the tool has already got the high baud rate.

5.11. RETRANSFER Feature

Once the Flashing operation completed, if the user wishes to reflash the same binaries for any reason, clicking on retransfer button will reset the tool and the user can flash the already selected files.

5.12. Closing the Flashing tool

Click on the close button to close the flashing tool as shown in figure-28.

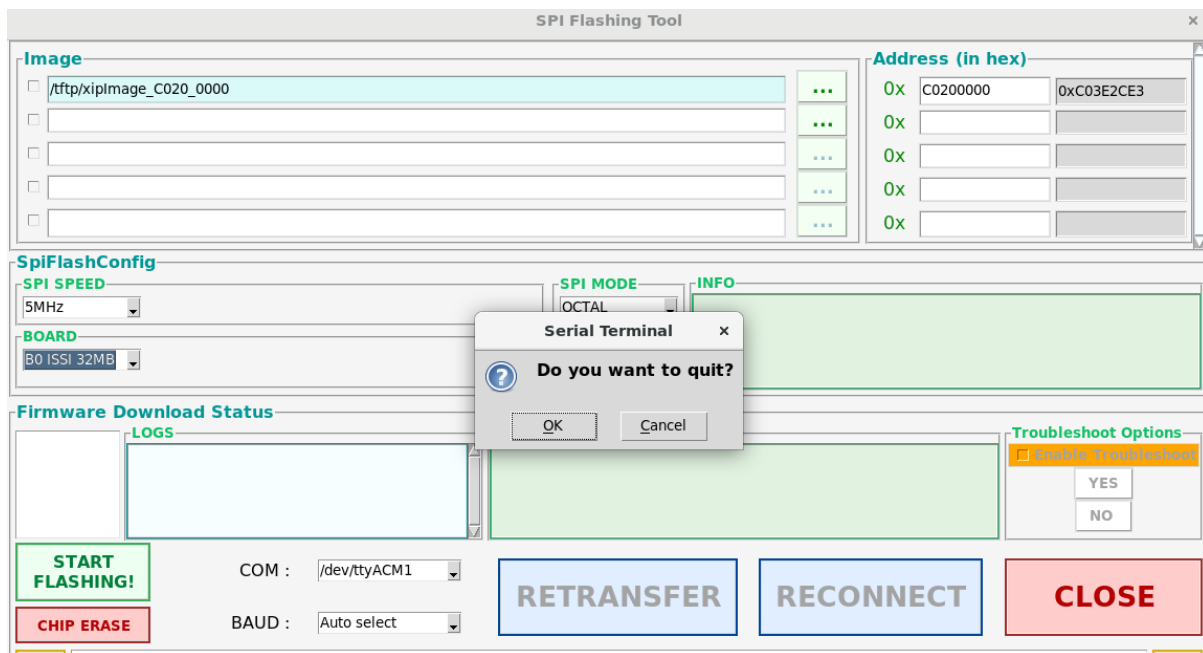


Figure 27: Quitting the tool

NOTE:

- During the operation, the CLOSE is disabled hence the user cannot close the tool. If the user tries to close the tool by clicking on 'x' button in taskbar, the tool will notify the user not to close since the operation is in progress as shown in Figure-29.

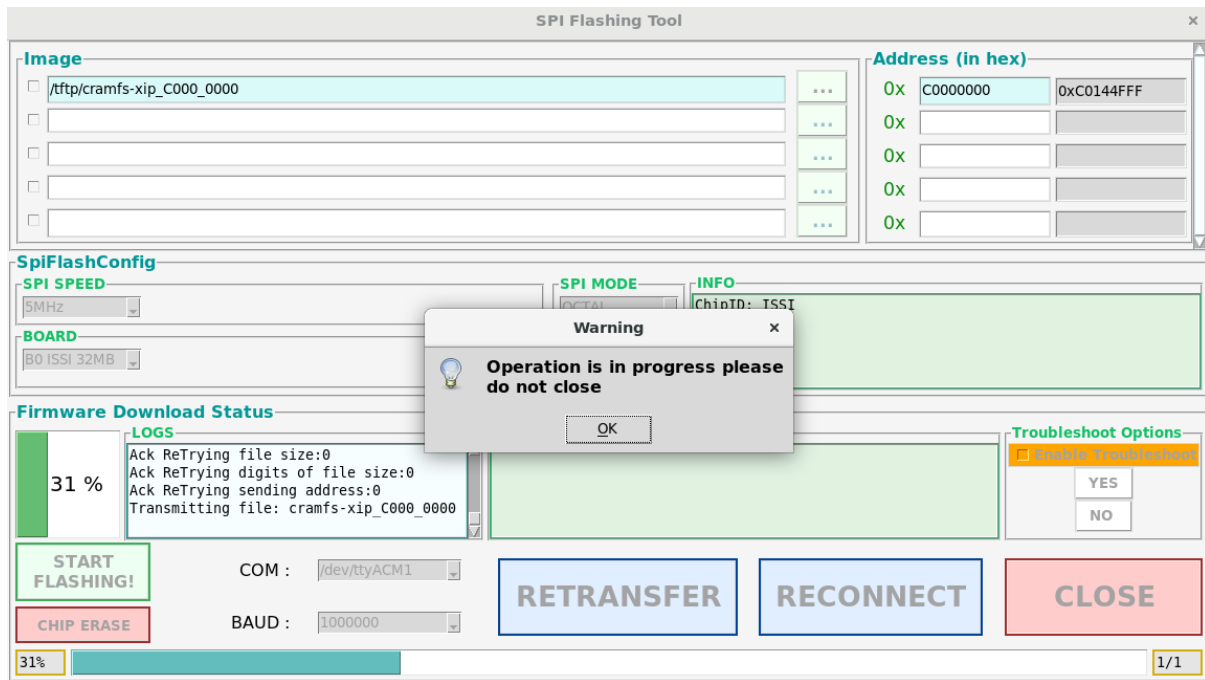


Figure 28: Trying to close the tool during operation

6. Using the Command line

This chapter demonstrates how to run the flashing tool through command line.

6.1. Requirements:

Python modules used in the implementing flashing tool are

- serial – (If not installed, run this command `pip install pyserial`)
- argparse – builtin
- sys – builtin
- os – builtin
- time – builtin
- threading – builtin
- libserial - user defined - libserial.py

libserial.py: All the methods related to serial including opening, reading and writing are written in libserial.py under SerialComm class.

NOTE: Before running the flashing tool, make sure that “.flash.cfg” file is provided in the same directory in which the flash_programming_cmdline.py file exists.

6.2. Usage:

- In order to perform flashing of file, type the below command in the terminal,

```
$ sudo python3 flash_programming_cmdline.py --board BOARD
--dfs DFS --COM COM --autoselectbaud AUTOSELECTBAUD
--spispeed SPISPEED --spimode SPIMODE --file1 FILE1
--address1 ADDRESS1 --file2 FILE2 --address2 ADDRESS2
--file3 FILE3 --address3 ADDRESS3 --file4 FILE4 --
address4 ADDRESS4 --file5 FILE5 --address5 ADDRESS5
```

Or

```
$ sudo python3 flash_programming_cmdline.py -b FLASH -d
DFS -c COM -asb AUTOSELECTBAUD -ss SPISPEED -sm SPIMODE
-f1 FILE1 -a1 ADDRESS1 -f2 FILE2 -a2 ADDRESS2 -f3 FILE3
-a3 ADDRESS3 -f4 FILE4 -a4 ADDRESS4 -f5 FILE5 -a5 ADDRESS5
```

- To perform chip erase, type the below command in the terminal

```
$ sudo python3 flash_programming_cmdline.py --board
FLASH -dfs DFS --COM COM --autoselectbaud AUTOSELECTBAUD
--spispeed SPISPEED --spimode SPIMODE --chiperase=True
```

Or

```
$ sudo python3 flash_programming_cmdline.py -b BOARD -d  
DFS -c COM -asb AUTOSELECTBAUD-ss SPISPEED -sm SPIMODE -  
cr True
```

Description of each option

- --board, -b: Board name as per the .flash.cfg.xml file
- --dfs, -d: Data frame size. 32
- --COM, -c: Serial port to which target is connected. Eg - /dev/ttyUSB0, COM3, COM5 etc
- --autoselectbaud, -asb: Auto select the baud rate at which file has to be transmitted
- --spispeed, -ss: 5MHz or 10MHz or 20MHz or 30MHz
- --spimode, -sm: OCTAL (only one option)
- --file, -f: file that need to be transferred
- --address, -a: flash address at which file has to be transferred
- --chiperase, -cr: perform chip erase, use chiperase=True

- Example

- \$ sudo python3 flash_programming_cmdline.py -b "A1 ISSI 32MB" -d 32 -c /dev/ttyUSB1 -asb 1 -ss 10MHz -sm OCTAL -cr True

```

shaik@alif:~/developmentstudio-workspace/host_local_repo$ sudo python3 flash_programming_cmdline.py -h
[sudo] password for shaik:
usage: flash_programming_cmdline.py [-h] -b BOARD -d DFS -c COM -asb
                                AUTOSELECTBAUD -ss SPISPEED -sm SPIMODE
                                [-f1 FILE1] [-f2 FILE2] [-f3 FILE3]
                                [-f4 FILE4] [-f5 FILE5] [-a1 ADDRESS1]
                                [-a2 ADDRESS2] [-a3 ADDRESS3]
                                [-a4 ADDRESS4] [-a5 ADDRESS5]
                                [-cr CHIPERASE]

optional arguments:
  -h, --help            show this help message and exit
  -b BOARD, --board BOARD
                        Mention the board type, DevBoard or CarrierBoard
  -d DFS, --dfs DFS     Mention the data frame size. Options - 32
  -c COM, --COM COM     Mention the port/device name
  -asb AUTOSELECTBAUD, --autoselectbaud AUTOSELECTBAUD
                        Auto select the baudrate. Options - 0,1
  -ss SPISPEED, --spispeed SPISPEED
                        Mention the SPI speed. Options - 5MHz, 10MHz and 20MHz
  -sm SPIMODE, --spinode SPIMODE
                        Mention the SPI mode. Options - OCTAL
  -f1 FILE1, --file1 FILE1
                        Specify the path of the first file you wish to
                        transfer
  -f2 FILE2, --file2 FILE2
                        Specify the path of the second file you wish to
                        transfer
  -f3 FILE3, --file3 FILE3
                        Specify the path of the third file you wish to
                        transfer
  -f4 FILE4, --file4 FILE4
                        Specify the path of the fourth file you wish to
                        transfer
  -f5 FILE5, --file5 FILE5
                        Specify the path of the fifth file you wish to
                        transfer
  -a1 ADDRESS1, --address1 ADDRESS1
                        Specify the address for the first file you wish to
                        transfer
  -a2 ADDRESS2, --address2 ADDRESS2
                        Specify the address for the second file you wish to
                        transfer
  -a3 ADDRESS3, --address3 ADDRESS3
                        Specify the address for the third file you wish to
                        transfer
  -a4 ADDRESS4, --address4 ADDRESS4
                        Specify the address for the fourth file you wish to
                        transfer
  -a5 ADDRESS5, --address5 ADDRESS5
                        Specify the address for the fifth file you wish to
                        transfer
  -cr CHIPERASE, --chiperase CHIPERASE
                        Chiperase

```

Figure 29: *flash_programming_cmdline.py* usage

6.3. Performing flashing:

Use the command for performing flashing as shown in usage section.

After executing the command, if ‘-asb’ is 0, baud rate 115200 is used. If ‘-asb’ is 1, the tool will try to switch to the higher baud rate upto 1000000 if available and then start communicating with the target and on successful handshake with the target, flashing of the file will begin as shown in the below figure.

```

shaik@alif:~/developmentstudio-workspace/host_local_repo$ sudo python3 flash_programming_cmdline.py -b CarrierBoard -d 32 -c /dev/ttyUSB0 -asb 1 -ss 10MHz -sm OCTAL -f1 /home/shaik/developmentstudio-workspace/version_1.3_release/binary_files/xipImage -a1 0x00000000 -f2 /home/shaik/developmentstudio-workspace/version_1.3_release/binary_files/cramfs -a2 0x02000000
..... FLASHING TOOL .....
Address Range of file1: 0x00000000 - 0x0B10E21F
Address Range of file2: 0x02000000 - 0x0B33777F
checking baudrate substitution

```

Figure 30: Checking baud rate switching

```
----- FLASHING TOOL -----  
  
Address Range of file1: 0XD8000000 - 0XD81DE21F  
Address Range of file2: 0XD8200000 - 0XD835FFFF  
Checking baudrate switching.....  
Baudrate : 1000000  
Serial port successfully opened  
Flash size : 0X400000  
Target Ready?  
Target is connected  
Chip ID : ADEST0  
Preparing flash....
```

Figure 31: Prepare flash

Once the flashing is done, the log will show as shown in Figure-34 and all the logs including handshake and other details are recorded in the Logsfiler.txt. Figure 33 shows the ongoing flashing process and Figure 34 represents the completed flashing process.

```
----- FLASHING TOOL -----  
  
Address Range of file1: 0XD8000000 - 0XD81DE21F  
Address Range of file2: 0XD8200000 - 0XD835FFFF  
Checking baudrate switching.....  
Baudrate : 1000000  
Serial port successfully opened  
Flash size : 0X400000  
Target Ready?  
Target is connected  
Chip ID : ADEST0  
Transmitting file : xipImage  
Flashing : [ ] 53%
```

Figure 32: Ongoing flash

```

----- FLASHING TOOL -----
Address Range of file1: 0XD8000000 - 0XD81DE21F
Address Range of file2: 0XD8200000 - 0XD835FFFF
Checking baudrate switching.....
Baudrate : 1000000
Serial port successfully opened
Flash size : 0X400000
Target Ready?
Target is connected
Chip ID : ADEST0
Transmitting file : xipImage
Flashing : [ ] 100%
Transferred xipImage file in 123.24167
Transmitting file : cramfs
Flashing : [ ] 100%
Transferred cramfs file in 90.98621

Transmission completed
#####Done in 214.22782 sec#####
Serial port closed successfully
-----

```

Figure 33: Flashing completed

6.4. Performing chip erase:

Use the command for performing chip erase as shown in usage section

After executing the command, the flashing tool asks the user whether he wants to continue flash erase or not as shown in the figure 35.

```

shaik@alif:~/developmentstudio-workspace/host_local_repo$ sudo python3 flash_programming_cmdline.py -b CarrierBoard -d 32 -c /dev/ttyUSB0 -asb 1 -ss 10MHz -sm OCTAL -cr True
----- FLASHING TOOL -----
You are about to perform chip erase. Do you want to continue(Y/N)? : y

```

Figure 34: Asking user if he wants to continue with chip erase

Upon clicking Y, the tool will start erasing the flash. After completing the flash erase successfully, the log will display as shown in Figure-37. Else it reports ‘Chip erase failed’ on failure.

```

----- FLASHING TOOL -----
You are about to perform chip erase. Do you want to continue(Y/N)? : y
Checking baudrate switching.....
Baudrate : 1000000
Flash size : 0X400000
Performing chip erase.....

```

Figure 35: Performing chip erase after “y” option

```

----- FLASHING TOOL -----
You are about to perform chip erase. Do you want to continue(Y/N)? : y
Checking baudrate switching.....
Baudrate : 1000000
Flash size : 0X400000
Chip erase done
#####Done in 150 sec#####
Serial port closed successfully
-----

```

Figure 36: Chip erase done

7. Basic UART Test

This chapter demonstrates the basic UART test through command line.

7.1. Requirements

- basic_uart2_test_binary.axf
- uart_test.py

If GUI or cmdline tool fails to communicate with the target use this binary to verify the UART2 connection.

Check the basic uart2 rtscs working with the basic_uart2_test_binary. Run the binary and use the below python script to verify data is transmitted and ack received properly. To confirm uart binary is working, on the terminal where python file with below python code is running b'OK\n' should be there as ack.

uart_test.py contains:

```
import serial
import time
ser = serial.Serial('/dev/ttyUSB0',115200,timeout=30,rtscs=True)
time.sleep(5)
ser.write(b'ABCDEFGHIJKLMNOPQRSTUVWXYZ012345ABCDEFGHIJKLMNOPQRSTUVWXYZ012345ABCDEFGHIJKLMNOPQRSTUVWXYZ012345ABCDEF
GHIJKLMNOPQRSTUVWXYZ012345ABCDEFGHIJKLMNOPQRSTUVWXYZ012345ABCDEFGHIJKLMNOPQRSTUVWXYZ012345ABCDEF
GHIJKLMNOPQRSTUVWXYZ012345ABCDEFGHIJKLMNOPQRSTUVWXYZ012345ABCDEFGHIJKLMNOPQRSTUVWXYZ012345')
print("Write completed")
s=ser.readline()
print(s)
ser.close()
```

Run the basic_uart2_test_binary.axf and uart_test.py, the following will be displayed on the host terminal

```
Write completed
OK
```

Appendix

The project zip file contains two directories/folders:

1. host_tool
2. target_binary

host tool:

1. This folder contains 3 files namely:
 - i. flashtool.py - Python code for Flashing utility GUI
 - ii. libserial.py - Python code for serial communication
 - iii. flash_programming_cmdline.py - Python code for flashing utility command line.
 - iv. .flash.cfg - File having flash and size information used by flashtool.py and flash_programming_cmdline.py
2. For system Prerequisite, installation steps and usage, please refer Chapter 2 and 3.

target binary:-

1. This folder contains target binary which needs to run on arm cortex m55 platform using trace32 debugger.
2. We have followed the steps provided by alif to use and launch this binary using trace32 debugger.
3. The target binary provided supports flashing for both dev board and carrier board for UART2 configuration.